



Nothing the Rain Hadn't Done: A Pre-Registered Stepped-Wedge Trial of Quarterly Rooftop-Solar Washing Across Forty Community Sports Clubs

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Abstract. Quarterly manual washing of rooftop photovoltaic arrays is a near-universal line item in community-facility maintenance schedules, and is ordinarily justified on generation grounds. We report a pre-registered stepped-wedge cluster trial across forty community sports clubs in a single metropolitan council's facilities portfolio, in which the quarterly wash was introduced in four randomised sequences over eight quarters. The primary outcome was the performance ratio derived from inverter energy and plane-of-array irradiance, with equivalence bounds of ± 1.5 percentage points fixed before the first wash. The estimated effect of a wash on performance ratio across the fourteen days following it was 0.11 pp (95% CI -0.42 to 0.64); two one-sided tests reject effects outside the pre-registered bounds ($p < 0.01$). The estimate is stable across every reduced specification considered, and no club's individual estimate falls outside the equivalence interval. Rainfall in the study region appears to account for the majority of surface recovery. We report the null as a null. Over the same period the wash was retained in the council's Asset Condition Attestation, having been reclassified from a generation-assurance activity to a condition-evidence activity, and the number of attested items per club rose from six to nine.

Introduction

Soiling — the accumulation of dust, pollen, salt, and biological deposit on a photovoltaic module's glazing — is a well-characterised cause of generation loss, and the literature quantifying it is mature [1], [2]. In large arid-climate plants the economic case for periodic cleaning is frequently decisive, and considerable effort has gone into scheduling that cleaning well [3], [4].

Community facilities are not large arid-climate plants. A suburban sports club's rooftop array is small, shallowly pitched, and sits under temperate rainfall; its maintenance is nonetheless procured, scheduled, and attested through the same apparatus that governs the rest of the facility. The quarterly wash arrives in the schedule with a generation rationale attached and is thereafter renewed without one, because the

schedule is the artefact the renewal process reads.

We take the rationale at face value and test it. Our contributions are, we suggest, threefold:

- a pre-registered stepped-wedge trial of the quarterly wash across a whole council portfolio, with equivalence bounds fixed in advance, which may offer a cleaner estimate than the before-and-after comparisons the sector currently relies on;
- an equivalence result rather than a failure to reject, which we believe supports the stronger claim that any effect present is smaller than the smallest effect the program was designed to secure;
- an account of what the maintenance program did with that result, which appears to generalise beyond the wash itself.

Background

The physical mechanisms of deposition and removal are understood in some detail, including the role of precipitation in restoring transmittance [2], and predictors of soiling loss have been investigated across a range of sites [5]. Techno-economic treatments weigh the recoverable yield against the cost of intervention [4], and scheduling models exist to place cleanings where that balance is most favourable [3]. The maintenance-policy literature more broadly has long distinguished interventions justified by the condition of a deteriorating system from those justified by the calendar [6].

A parallel literature concerns what happens to a procedure once it is institutionalised. Formal structures may be sustained as ceremonial displays that are only loosely coupled to the technical work they nominally serve [7], and organisations within a field converge on common practice irrespective of local efficiency [8]. The diffusion of explicit performance standards through public administration [9] has been shown to produce measures that reshape the activity they record [10], sometimes at the expense of the outcome the measure stands for [11].

Prior work at this institution bears on the first of these. A validation study of an occupancy-aware quad irrigation controller found no measurable relationship between reported occupancy and watering suppression, and no significant change in watering behaviour once the occupancy sensor was removed altogether [12]. A matched case-control study of public park barbecue units similarly found unit condition and recorded usage sharply divergent, on a counter that increments only when a unit completes a heat cycle [13]. Both describe a schedule or a record proceeding independently of the thing it is supposed to track.

Method

Design. A stepped-wedge cluster trial across forty community sports clubs holding rooftop arrays under one metropolitan council’s facilities portfolio. Clubs were randomised into four sequences of ten; each sequence crossed from no scheduled wash to a quarterly wash at a different quarter, so that by the eighth quarter all forty clubs had crossed. Randomisation was stratified by installed capacity band. The protocol, primary outcome, equivalence bounds, and

analysis model were registered before the first wash [14].

Wash protocol. Deionised-water rinse with a soft-bristle brush, applied by the council’s existing grounds contractor to the whole array, logged with a timestamped completion record. No detergent, no coating, no panel-level intervention.

Measurement. Fifteen-minute inverter energy and plane-of-array irradiance from sensors installed at each site twelve months before the trial opened. The primary outcome is the performance ratio, computed for site i in interval t as

$$PR_{it} = \frac{E_{it}}{P_i \cdot \left(\frac{H_{it}}{G_{STC}} \right)}$$

where E_{it} is metered AC energy, P_i the installed DC capacity, H_{it} the plane-of-array insolation, and G_{STC} the reference irradiance. Intervals with curtailment, inverter fault, or irradiance below 50 W m^{-2} were excluded before analysis under the registered rule.

Analysis. A linear mixed model with site and quarter fixed effects and a site random intercept, estimated over the fourteen days following each wash against the matched pre-wash baseline, with cluster-robust standard errors to account for serial correlation within site [15]. Staggered adoption was handled with a group-time treatment-effect estimator [16]. Daily rainfall from the nearest gauge entered as a covariate, and a soiling-season interaction was pre-specified. Equivalence against the registered ± 1.5 pp bounds was assessed by two one-sided tests [17], [18].

Results

Forty clubs contributed 312 club-quarters and 1,184 logged washes. Site characteristics are given in Table 1.

Band	Clubs	Median kWp	Pitch
< 10 kWp	14	6.6	8°
10–30 kWp	15	19.2	11°
> 30 kWp	11	44.8	6°

Table 1: Portfolio composition. No band contains fewer than eleven clubs, and the three medians span the range the council’s own procurement schedule treats as a single class.

The two arms track one another throughout the measurement window (Figure 1). Both fall through the dry late-summer weeks and both

recover with the first sustained rain, with the washed arm ahead in thirteen of twenty-six weeks and behind in the remainder.

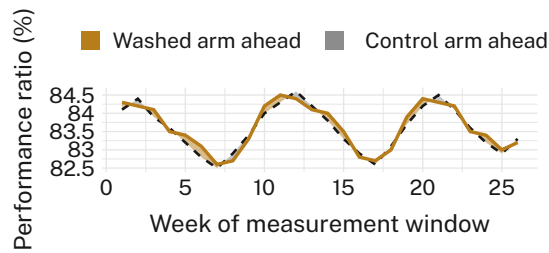


Figure 1: Weekly performance ratio by arm. The two series cross fourteen times across the window, and the largest weekly separation between them is 0.3 pp – smaller than the seasonal swing either arm undergoes between weeks 12 and 17.

The pre-registered estimate of the wash effect on performance ratio over the following fourteen days is 0.11 pp (95% CI –0.42 to 0.64). Both one-sided tests against the ± 1.5 pp bounds reject ($p < 0.01$), so the interval is contained within the registered equivalence region. The twenty-eight-day window returns 0.09 pp (95% CI –0.39 to 0.57).

Per-club estimates are shown in Figure 2. The dispersion across clubs is appreciable relative to the point estimate and negligible relative to the equivalence bounds.

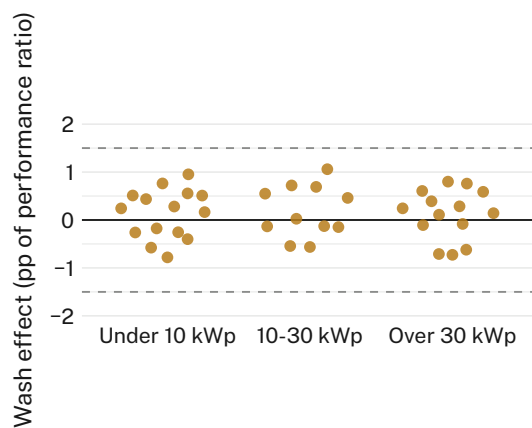


Figure 2: Per-club wash effect, one point per club, against the registered ± 1.5 pp bounds (dashed). All forty club-level estimates fall inside the bounds; the largest, at 1.05 pp, belongs to a club in the middle capacity band.

Rainfall is the dominant term. Days following a rainfall event above 4 mm are associated with a recovery of 1.9 pp (95% CI 1.4 to 2.4) irrespective

of arm, which is an order of magnitude above the wash estimate and the reason we regard the study region as the operative condition rather than a nuisance.

The pre-registered specification was compared against three reduced ones (Figure 3). Removing the site random intercept moves the estimate by 0.03 pp (n.s.); removing the rainfall covariate moves it by 0.03 pp in the other direction (n.s.); dropping the soiling-season interaction moves it by 0.02 pp (n.s.). We retain all three terms for completeness.

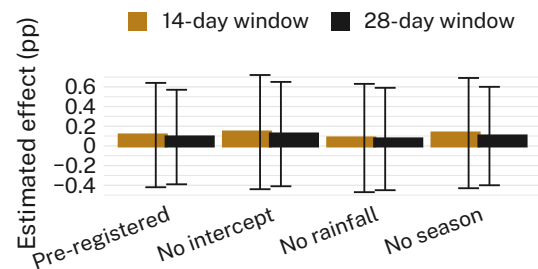


Figure 3: Ablation across four specifications and two accumulation windows. No pair of the eight estimates differs by more than 0.07 pp, and every interval covers zero.

Over the same eight quarters the wash remained on the council's Asset Condition Attestation. Following the interim analysis at quarter six it was reclassified from a generation-assurance activity to a condition-evidence activity, on the grounds that a completed wash establishes that the roof was accessed and the array visually inspected. Attested items per club rose from six to nine across the trial (Figure 4).

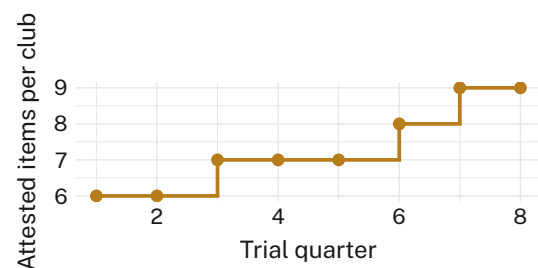


Figure 4: Attested maintenance items per club by trial quarter. The count rises at quarters 3, 6, and 7; no item was retired during the trial period, and quarterly attestation completeness held at 100% throughout.

Discussion

We observe consistent equivalence across settings, windows, and specifications. The wash

does not appear to move the performance ratio at any magnitude the program was established to secure, and the surface recovery the program was designed to deliver appears to be delivered instead by rainfall. We frame this as an equivalence result rather than an absence of evidence, since the pre-registered bounds make the claim a positive one [18]; the effect, if present, is smaller than the study was built to care about [19].

That the schedule survived the finding is, on the institutional reading, the expected outcome rather than an anomalous one. A practice sustained because it generates evidence of its own completion is loosely coupled to the technical account that introduced it [7], and the reclassification at quarter six preserved the practice by relocating its rationale to a claim no generation measurement can bear on. We note this without prejudice to the reclassification, which is administratively sound.

Limitations

The trial covers a single council portfolio in a single temperate rainfall regime, though the consistency of the estimate across all three capacity bands and both accumulation windows suggests the result may travel to comparable portfolios. Our measurement window excludes the two months of heaviest pollen load, which would if anything favour the washed arm and did not. We did not randomise the reclassification decision, and are unable to say whether an alternative rationale would have been located had this one been unavailable; we consider it likely that one would.

Conclusion

A pre-registered stepped-wedge trial across forty community sports clubs finds the quarterly rooftop-solar wash equivalent to no wash within bounds fixed before the first intervention, with rainfall accounting for the recovery the wash was procured to provide. We report the null as a null, in the view that the literature's tolerance for such reports remains lower than is useful [20], [21]. Future work might follow the reclassified item through a further attestation cycle, and establish whether an activity that has changed rationale once can be observed to do so again.

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